VANISHING OF A PRO-SYSTEM IS EQUIVALENT TO THE VANISHING OF THE LIMIT AND FIRST DERIVED LIMIT OF IT AND OF ITS COUNTABLE DIRECT SUM

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Editorial note. The statement's second condition refers to the properties also holding for the countable direct sum of (A_n), delivered verbatim. The two references inside the counted proof span are recorded as non-load-bearing navigation pointers.

Allowed standard prerequisites (editorial).

- Pro-objects, pro-isomorphisms and the notion of a system being zero as a pro-object. used at statement line 23531 and proof lines 23543-23544. Is one of the two conditions being compared.
- Emmanouil's criterion for an inverse system to be Mittag-Leffler. invoked at proof line 23540. Converts the vanishing hypotheses into Mittag-Lefflerness.
- Surjectivity of the maps A_{n+1,m_{n+1}} -> A_{n,m} in the Mittag-Leffler case. used at proof lines 23547-23548. Is what lets the vanishing of the limit descend to the images.

Source reference. Original lemma 23422: sources/stacks-more-algebra/pinned-original-source.tex, source lines 23422–23422. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 23073: sources/stacks-more-algebra/pinned-original-source.tex, source lines 23073–23073. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Lemma 0.2. *Let (A_n) be an inverse system of abelian groups. The following are equivalent*

(1) *(A_n) is zero as a pro-object,*
(2) *$\lim A_n = 0$ and $R^1 \lim A_n = 0$ and the same holds for $\bigoplus_{i \in \mathbf{N}} (A_n)$.*

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Proof. It follows from Lemma 23073 that (1) implies (2). Assume (2). Then (A_n) is ML by Lemma 23422. For $m \geq n$ let $A_{n,m} = \mathrm{Im}(A_m \to A_n)$ so that $A_n = A_{n,n} \supset A_{n,n+1} \supset \ldots$. Note that (A_n) is zero as a pro-object if and only if for every n there is an $m \geq n$ such that $A_{n,m} = 0$. Note that (A_n) is ML if and only if for every n there is an $m_n \geq n$ such that $A_{n,m} = A_{n,m+1} = \ldots$. In the ML case it is clear that $\lim A_n = 0$ implies that $A_{n,m_n} = 0$ because the maps $A_{n+1,m_{n+1}} \to A_{n,m}$ are surjective. This finishes the proof. $\square$